CHAPTER 5

CHEMICAL THERMODYNAMICS

CHAPTER ESSENCE AT A GLANCE:
Governs energy transformations, enthalpy changes, Hess's Law of heat summation, entropy, and Gibbs free energy criteria for spontaneity.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **First Law of Thermodynamics:** Energy is conserved; change in internal energy equals heat absorbed plus work done on the system: Delta U = q + w.

• **Enthalpy & Heat Capacity:** Enthalpy H = U + pV measures constant-pressure heat; molar heat capacities satisfy Mayer's relation Cp - Cv = R.

• **Hess's Law of Constant Heat Summation:** Total enthalpy change in a reaction is independent of whether it occurs in one step or multiple steps.

• **Gibbs Free Energy & Spontaneity:** A process is spontaneous at constant T and P if and only if the change in Gibbs free energy is negative: Delta G < 0.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\Delta U = q + w = q - p_{\text{ext}}\Delta V$$

First Law of Thermodynamics (IUPAC convention)

$$\Delta H = \Delta U + \Delta n_g RT$$

Enthalpy and Internal Energy Relation for Gases

$$\Delta_r H^\circ = \sum \Delta_f H^\circ(\text{products}) - \sum \Delta_f H^\circ(\text{reactants})$$

Standard Enthalpy of Reaction via Formation Enthalpies

$$\Delta G = \Delta H - T\Delta S \quad \text{and} \quad \Delta G^\circ = -2.303RT\log K$$

Gibbs-Helmholtz Equation and Equilibrium Constant

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In Delta H = Delta U + Delta n_g RT, Delta n_g counts ONLY moles of gaseous products minus gaseous reactants.

• **Alert:** An endothermic reaction (Delta H > 0) can still be spontaneous at high temperatures if entropy change Delta S is positive and T Delta S > Delta H.

• **Alert:** Standard enthalpy of formation for elements in their standard reference states (e.g., O2 gas, C graphite) is identically zero.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the First Law of Thermodynamics and write its mathematical equation. Define a state function. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Statement and Formulation:** Energy can neither be created nor destroyed; total energy of the universe remains constant. Mathematically: Delta U = q + w, where Delta U is internal energy change, q is heat absorbed, and w is work done. **[1 Mark]**

• **State Function:** A thermodynamic property whose value depends solely on the current state of the system, completely independent of the path taken (e.g., U, H, S, G, T, P). **[1 Mark]**

• **Key Exam Result:** Internal energy U is a state function; heat q and work w are path functions.

2. Define standard enthalpy of formation. What is Delta_f H° for elemental O2 (g) and C (graphite)? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The enthalpy change accompanying the formation of one mole of a compound from its constituent elements in their most stable standard states at 298 K and 1 bar pressure. **[1 Mark]**

• **Elemental Reference Enthalpy:** By convention, standard enthalpy of formation of every pure element in its reference standard state is strictly ZERO. Therefore, Delta_f H° = 0 for O2 (g) and C (graphite). **[1 Mark]**

• **Key Exam Result:** Standard enthalpies of formation of pure reference elements are assigned zero by convention.

3. Derive the mathematical relation $\Delta H = \Delta U + \Delta n_g R T$ for a chemical reaction involving ideal gases. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Enthalpy Definition):** Enthalpy is defined as H = U + pV. For a finite change at constant pressure: $\Delta H = \Delta U + p \Delta V = \Delta U + p(V_B - V_A)$. **[1 Mark]**

• **Step 2 (Ideal Gas Law Substitution):** Assuming ideal gas behaviour for reactants and products: $p V_A = n_A R T$ and $p V_B = n_B R T$. Hence, $p(V_B - V_A) = (n_B - n_A) R T$. **[1 Mark]**

• **Step 3 (Final Expression):** Let $\Delta n_g = n_B(g) - n_A(g)$, the difference in moles of gaseous products and gaseous reactants. Therefore: $\Delta H = \Delta U + \Delta n_g R T$. **[1 Mark]**

• **Key Exam Result:** If $\Delta n_g = 0$, ΔH equals ΔU; if $\Delta n_g > 0$, ΔH exceeds ΔU.

$$\Delta H = \Delta U + \Delta n_g RT \quad (\Delta n_g = n_{\text{gas, prod}} - n_{\text{gas, react}})$$

Enthalpy-Internal Energy Gas Equation

4. State Hess's Law of Constant Heat Summation. Illustrate with an example showing that reaction enthalpy is independent of the path. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Statement:** The total enthalpy change for a chemical reaction is identical whether the reaction takes place in one single step or in a series of multiple intermediate steps. **[1 Mark]**

• **Illustrative Example (Carbon Combustion):** Direct path: C (graphite) + O2 (g) → CO2 (g), ΔH = -393.5 kJ/mol. **[1 Mark]**

• **Two-step Alternative Path:** Step 1: C + 0.5 O2 → CO, $\Delta H1$ = -110.5 kJ/mol. Step 2: CO + 0.5 O2 → CO2, $\Delta H2$ = -283.0 kJ/mol. Total $\Delta H = \Delta H1 + \Delta H2$ = -110.5 - 283.0 = -393.5 kJ/mol. **[1 Mark]**

• **Key Exam Result:** Hess's law enables determination of reaction enthalpies that cannot be measured directly in calorimetry.

5. State the thermodynamic criteria for spontaneity of a process in terms of Delta G, Delta H, and Delta S. Under what conditions does an endothermic reaction become spontaneous? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Gibbs Spontaneity Equation:** Delta G = Delta H - T Delta S. A process is spontaneous if Delta G < 0, non-spontaneous if Delta G > 0, and at dynamic equilibrium if Delta G = 0. **[1 Mark]**

• **Endothermic Spontaneity Criteria:** For an endothermic reaction, Delta H > 0. For Delta G to be negative, Delta S must be positive (disorder increases) and the temperature T must be sufficiently high such that T Delta S > Delta H. **[1 Mark]**

• **Threshold Temperature:** Spontaneity begins at temperatures T > Delta H divided by Delta S. **[1 Mark]**

• **Key Exam Result:** Endothermic reactions become spontaneous only at high temperatures driven by positive entropy.

$$\Delta G = \Delta H - T\Delta S < 0 \rightarrow T > \frac{\Delta H}{\Delta S} \quad (\text{when } \Delta H > 0,\ \Delta S > 0)$$

Spontaneity Condition for Endothermic Process

6. For a chemical reaction, ΔH = 400 kJ/mol and ΔS = 0.2 $\frac{kJ}{K mol}$. Calculate the minimum temperature above which the reaction becomes spontaneous. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Spontaneity Condition):** The reaction is at equilibrium when Delta G = 0. That is, Delta H - T Delta S = 0. **[1 Mark]**

• **Step 2 (Temperature Calculation):** T = ΔH divided by ΔS = (400 kJ/mol) divided by (0.2 $\frac{kJ}{K mol}$) = 2000 K. **[1 Mark]**

• **Step 3 (Conclusion & Boxed Result):** At T = 2000 K, Delta G = 0. For any temperature higher than 2000 K, T Delta S exceeds Delta H, making Delta G negative and the process spontaneous. **[1 Mark]**

• **Key Exam Result:** The reaction becomes spontaneous at all temperatures above 2000 K.

$$T_{\text{equilibrium}} = \frac{\Delta H}{\Delta S} = \frac{400\ \text{kJ/mol}}{0.2\ \text{kJ/(K} \cdot \text{mol)}} = 2000\ \text{K}$$

Equilibrium Temperature Evaluation